DEEPAK BHAT

Software Engineer

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EDUCATION

CMR Institute of Technology, Bengaluru 2023 – 2027 (Expected)

Bachelor of Engineering in Computer Science and Engineering CGPA: 8.72 / 10

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Systems Design, Database Design, Operating Systems, Computer Networks, Scalable System Architecture, Machine Learning Fundamentals.

PROJECTS

OralCare – Dental Clinic Management Platform Spring Boot, React, PostgreSQL, REST APIs, Maven

- Architected scalable clinical management system with patient queue management and real-time tracking; engineered PostgreSQL schema and JWT/RBAC authentication reducing scheduling overhead by 35%.
- Designed and implemented 10+ REST APIs applying clean architecture principles; optimized PostgreSQL through indexing and query analysis, achieving sub-200ms response times with 99.2% system reliability.
- Developed admin dashboard automating critical workflows including email alerts, PDF report generation, and treatment tracking; eliminated 40+ manual hours monthly.
- Engineered real-time queue prediction system; analyzed N+1 query patterns and implemented database indexing strategy, scaling throughput by 45% for concurrent patient sessions.

BookMySlot – Event and Slot Booking Platform Spring Boot, React.js, MySQL, JWT, REST APIs

- Analyzed business requirements and designed normalized schema for event platform supporting 1,000+ concurrent users with 50+ daily bookings, demonstrating robust systems design.
- Engineered 15+ secure REST APIs with JWT/RBAC across four roles; implemented multi-level approval workflows and seamless event-to-slot associations.
- Optimized database indexing and schema normalization, delivering 30% query performance improvement and reducing page load times by 2.1 seconds.
- Built asynchronous notification pipeline with real-time alerts and interactive analytics dashboards; increased user engagement and retention by 35%.

FedMed-AI – Federated Learning Medical Diagnosis System Spring Boot, Python FastAPI, PostgreSQL, Docker, Nginx

- Architected distributed platform integrating Spring Boot and FastAPI for multi-disease risk analytics; processed 50,000+ patient records across decentralized hospital sources securely.
- Designed resilient asynchronous job processing using HTTP 202 polling and Resilience4j circuit breakers; sustained 99.9% uptime while handling 10x traffic spikes.
- Implemented HIPAA-compliant security with AOP audit logging, PHI masking, JWT refresh-token rotation, and rate limiting; achieved 100% audit pass rate.
- Deployed multi-stage Docker containers behind Nginx reverse proxy with Flyway migrations and JPA EntityGraphs; cut deployment time by 50%.

TECHNICAL SKILLS

Programming Languages:	Java, JavaScript, SQL, Python, C
Software Engineering:	REST APIs, API Design, Microservices Architecture, Systems Design, Software Architecture
Backend Frameworks:	Spring Boot, Spring Framework, Spring Security, JWT Auth, Authorization, FastAPI
Frontend Development:	React.js, HTML, CSS, Responsive Design
Database Management:	MySQL, PostgreSQL, Database Design, Query Optimization, Indexing, Schema Design
Tools & Platforms:	Git, Maven, Docker, Nginx, Swagger, OpenAPI, Postman, JPA/Hibernate, Resilience4j

LEADERSHIP & ACHIEVEMENTS

- 1st Place, CSE Department Project Competition (2025): Won against 50+ department entries for developing OralCare; demonstrated systems design and database design expertise handling 200+ daily transactions.
- Team Lead, Smart India Hackathon 2025: Led 6-member cross-functional team in collaborative environment; designed and deployed water-borne disease tracking solution emphasizing teamwork and scalable architecture for college selection.
- Runner-Up, DSA Hackathon, CMRIT: Placed 2nd among 100+ competitors, demonstrating analytical thinking, data structure optimization, and algorithmic problem-solving under time constraints.